

EEE554 – Probability and Random Processes

Module 8: Expectation

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Why Expectation?

Many engineering problems require computing an average.

Examples:

- Average packet delay
- Average received power
- Average queue length
- Average battery lifetime

How do we compute the average value of a random variable?

Expectation

The expectation (or mean) of a random variable is its probability-weighted average.

For a continuous random variable,

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

For a discrete random variable,

$$E[X] = \sum_x x p_X(x).$$

Why Does it Take This Form?

Suppose that X is observed repeatedly. Let

$$X_1, X_2, \dots, X_N$$

be N independent realizations of X , taking values from the finite discrete set $\{v_1, \dots, v_K\}$.

The sample average is

$$\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i.$$

Suppose that N_k observations are equal to v_k . Then

$$\bar{X}_N = \frac{1}{N} (N_1 \cancel{v}_1 + N_2 \cancel{v}_2 + \cdots) = \sum_k \cancel{v}_k \frac{N_k}{N}.$$

As $N \rightarrow \infty$, the relative frequencies satisfy

$$\frac{N_k}{N} \longrightarrow P(X = \cancel{v}_k).$$

Therefore,

$$\boxed{\bar{X}_N \longrightarrow \sum_k \cancel{v}_k P(X = x_k) = E[X].}$$

Remark: We will later formalize the convergence $\bar{X}_N \longrightarrow E[X]$ through the *Law of Large Numbers*.

Interpretation of Expectation

The expectation need not be a possible value of the random variable.

For example, suppose

$$X \sim \mathcal{U}\{1, \dots, 6\}.$$

Then

$$E[X] = \sum_{k=1}^6 k \frac{1}{6} = \frac{7}{2}.$$

However,

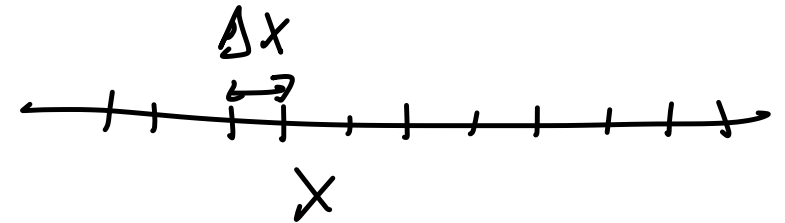
$$P\left(X = \frac{7}{2}\right) = 0.$$

The expectation describes the *center* of the probability distribution, not necessarily a realizable value of X .

Continuous Random Variables

A similar interpretation holds for continuous random variables. Partition the real line into small intervals

$$[x_i, x_i + \Delta x].$$



Suppose N_i observations fall into the i -th interval. Then

$$\bar{X}_N \approx \sum_i x_i \frac{N_i}{N}.$$

As $N \rightarrow \infty$, the relative frequencies satisfy

$$\frac{N_i}{N} \approx f_X(x_i) \Delta x.$$

Hence,

$$\bar{X}_N \approx \sum_i x_i f_X(x_i) \Delta x.$$

Taking $\Delta x \rightarrow 0$, yields

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

Interpretation

Expectation is the *center of mass* of the probability distribution.

Values with larger probabilities contribute more heavily to the average.

Expectation is the value around which the sample average converges as the number of observations becomes very large.

Example

Suppose $X \sim \mathcal{U}\{1, \dots, n\}$.

Then

$$E[X] = \sum_{k=1}^n k \frac{1}{n} = \frac{n+1}{2}.$$

Expectation of a Function

Suppose $Y = g(X)$.

How can we compute $E[g(X)]$?

For a continuous random variable,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

For a discrete random variable,

$$E[g(X)] = \sum_x g(x) p_X(x).$$

Other approach

calculus:

$f_Y(y)$

$$E(Y) = \int y \cdot f_Y(y) dy$$

Why this form?

Suppose we observe

$$X_1, X_2, \dots, X_N,$$

independent realizations of $X \in \{v_1, \dots, v_K\}$ The corresponding realizations of

$$Y = g(X)$$

are

$$g(X_1), g(X_2), \dots, g(X_N).$$

The sample average is

$$\frac{1}{N} \sum_{i=1}^N g(X_i).$$

Suppose N_k observations are equal to v_k . Then

$$\frac{1}{N} \sum_{i=1}^N g(X_i) = \sum_k g(v_k) \frac{N_k}{N}.$$

As $N \rightarrow \infty$,

$$\frac{N_k}{N} \longrightarrow P(X = v_k).$$

Hence,

$$E[g(X)] = \sum_k g(v_k) P(X = v_k).$$

Why is this Useful?

Suppose

$$Y = g(X).$$

One approach is to first determine the distribution of Y :

$$f_X \longrightarrow f_Y \longrightarrow E[Y].$$

However, if we are interested only in the expectation of Y , we can compute it directly:

$$E[Y] = E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

Thus, we do not need to determine the distribution of Y .

Example

Suppose

$$Y = X^2.$$

In the previous module, we determined the distribution of Y from the distribution of X .

If we are interested only in the average value of Y , this is unnecessary:

$$E[Y] = E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx.$$

Other Examples

$$g(x) = x \Rightarrow E[g(X)] = E[X].$$

$$g(x) = x^3 \Rightarrow E[g(X)] = E[X^3].$$

$$g(x) = |x| \Rightarrow E[g(X)] = E[|X|].$$

Probability as an Expectation

Let

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A, \end{cases}$$

denote the indicator function of the event A . Then

$$\boxed{P(X \in A) = E[\mathbf{1}_A(X)]}.$$

$$\begin{aligned} &= P(X \in A) \\ &= \int_A f(x) \cdot 1 \, dx + \int_{A^c} f(x) \cdot 0 \, dx \\ &= \int_{-\infty}^{\infty} f(x) \mathbf{1}_A(x) \, dx \end{aligned}$$

Interpretation

Probability is simply the expectation of an indicator random variable.

Expectation is therefore a generalization of probability.

Linearity

Expectation is linear.

$$= E(a \cdot X) + E(b \cdot Y)$$

$$E[aX + bY] = aE[X] + bE[Y].$$

More generally,

$$E \left[\sum_i a_i X_i \right] = \sum_i a_i E[X_i].$$

Important: Linearity of expectation does *not* require the random variables $X_1, \dots, X_n$ to be independent.

Counting Using Indicator Random Variables

Suppose I_i indicates whether event A_i occurs: $I_i = \mathbf{1}_{A_i}$.

Then

$$N = \sum_{i=1}^n I_i$$

counts the number of events that occur.

Using linearity of expectation,

$$E[N] = \sum_{i=1}^n E[I_i] = \sum_{i=1}^n P(A_i).$$

Important: This result does not require the events $A_1, \dots, A_n$ to be independent.

Conditional Expectation Given an Event

Given an event M , the conditional expectation is (continuous case)

$$E[X|M] = \int x f_{X|M}(x) dx.$$

and (discrete case)

$$E[X|M] = \sum_k x_k P_{X|M}(x_k)$$

$$f(x) = \lambda \cdot e^{-\lambda x}, x \geq 0$$

Example

Suppose $X \sim \text{Exp}(\lambda)$ models the waiting time in a queue (e.g., internet traffic). We find that

$$E[X] = \frac{1}{\lambda}, \quad E[X \mid X > x] = x + \frac{1}{\lambda}.$$

Hence, after already waiting x , the expected *additional* waiting time is

$$E[X - x \mid X > x] = \frac{1}{\lambda}.$$

This is a consequence of the *memoryless property* of the exponential distribution.

$$E(X) = \int_0^{+\infty} x \cdot \lambda e^{-\lambda x} dx = \int_0^{+\infty} x d e^{-\lambda x}$$

$$= -e^{-\lambda x} \cdot x \Big|_0^{+\infty} + \int_0^{+\infty} e^{-\lambda x} dx$$

$$= 0 - \frac{1}{\lambda} e^{-\lambda x} \Big|_0^{+\infty} = \boxed{\frac{1}{\lambda}}$$

$$M = \{X > x\}$$



$$F_{X|M}(t) = P(X \leq t | X > x)$$

$$= \frac{P(X \leq t \cap X > x)}{P(X > x)} = \begin{cases} 0, & t < x \\ \frac{P(X \in (x, t])}{P(X > x)}, & t \geq x \end{cases}$$

$$\underline{t \geq x} \quad P(X > x) = e^{-\lambda x} = 1 - F_x(x)$$

$$F_{x|M}(t) = \frac{F_x(t) - F_x(x)}{1 - F_x(x)}$$

$$= \frac{e^{-\lambda x} - e^{-\lambda t}}{e^{-\lambda x}} = 1 - e^{-\lambda(t-x)}$$

$$\Rightarrow f_{x|M}(t) = \lambda e^{-\lambda(t-x)}$$

$$M = X > x$$

$$E_{X|M}(X) = \int_{-\infty}^{+\infty} t \cdot f_{X|M}(t) \cdot dt$$

$$= \int_{-\infty}^{+\infty} t \cdot \lambda \cdot e^{-\lambda(t-x)} dt = x + \frac{1}{\lambda}$$

$t' = t - x$

$$= \int_0^{+\infty} (t' + x) \cdot \lambda \cdot e^{-\lambda t'} dt' = \underbrace{\int_0^{+\infty} t' \lambda e^{-\lambda t'} dt'}_{E(X)} + x \int_0^{+\infty} \lambda e^{-\lambda t'} dt'$$

Variance

Expectation measures the average.

Variance measures the spread around the average.

$$\boxed{\text{Var}(X) = E[(X - \mu)^2],}$$

where

$$\mu = E[X].$$

Spread case: $\text{Var}(X) = 0 \Rightarrow X = E(X)$ always!

$$E((X - \mu)^2) = E(X^2) + E(\mu^2) - 2\mu E(X)$$

Computing the Variance

$$= E(X^2) + \mu^2 - 2\mu \cdot \mu$$

Using linearity,

$$\boxed{\text{Var}(X) = E[X^2] - (E[X])^2.}$$

Standard Deviation

The standard deviation is

$$\boxed{\sigma = \sqrt{\text{Var}(X).}}$$

It has the same physical units as X .



Properties of Variance

For any random variable X ,

$$\text{Var}(X) \geq 0.$$

For constants $a, b \in \mathbb{R}$,

$$\boxed{\text{Var}(aX + b) = a^2 \text{Var}(X).}$$

$$\begin{aligned} Y &= aX + b \\ E(Y) &= a \cdot E(X) + b \\ \Rightarrow E\left[(Y - E(Y))^2\right] &= E\left[a^2 (X - E(X))^2\right] \\ &= a^2 E\left[(X - E(X))^2\right] \end{aligned}$$

In particular,

$$\text{Var}(X + b) = \text{Var}(X),$$

since adding a constant changes the mean but does not change the

spread.

Compare:

$$E[aX + b] = aE[X] + b,$$

whereas

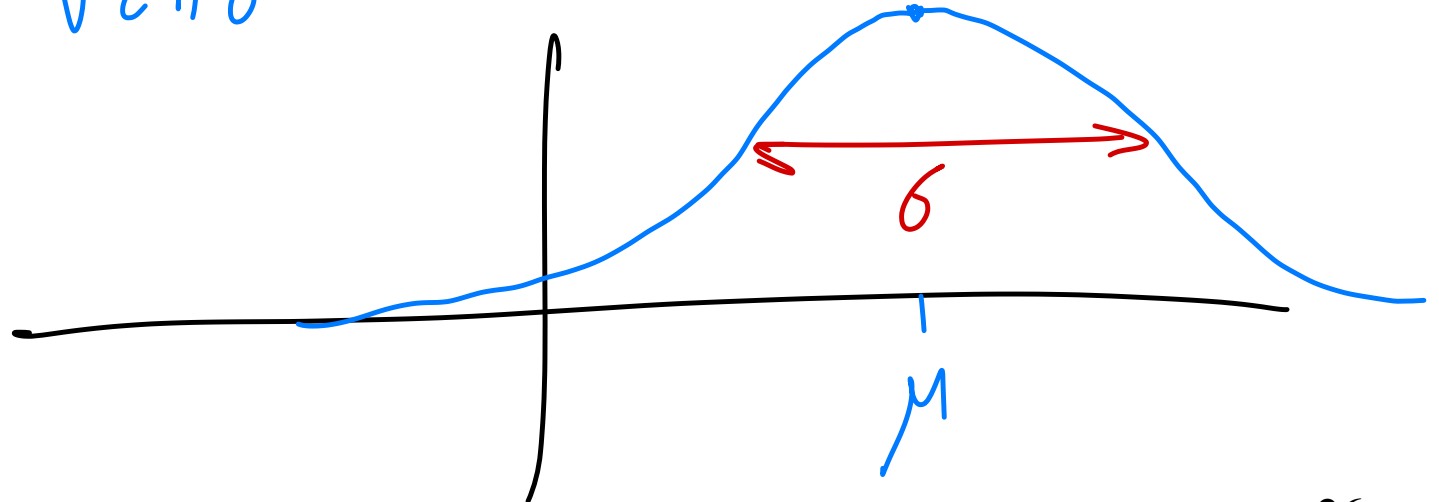
$$Var(aX + b) = a^2 Var(X).$$

Examples

Gaussian $X \sim \mathcal{N}(\mu, \sigma^2)$

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Poisson $X \sim \text{Poisson}(\lambda)$

$$E[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

$$\begin{aligned} E(X^2) &= (E(X))^2 + \text{Var}(X) \\ &= \lambda^2 + \lambda \end{aligned}$$

Exponential $X \sim \text{Exp}(\lambda)$

$$E[X] = 1/\lambda, \quad \text{Var}(X) = 1/\lambda^2.$$

Bernoulli $X \sim \text{Bernoulli}(p)$

$$E[X] = p, \quad \text{Var}(X) = p(1 - p).$$

$$X = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$$

$$E(X) = 1 \cdot p$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 \\ &= p(1-p) \end{aligned}$$

Physical Units

Suppose X represents a delay measured in seconds.

Then

$E[X]$ has units of seconds,

while

$Var(X)$ has units of seconds².

The standard deviation

$$\sigma_X = \sqrt{Var(X)}$$

therefore has the same physical units as X .

This is one reason standard deviation is often easier to interpret physically than variance.

Moments

The n -th raw moment is

$$E[X^n].$$

The n -th central moment is

$$E[(X - E[X])^n].$$

Examples

$$E[X]$$

(first moment)

$$E[X^2]$$

(second raw moment)

$$Var(X) = E[(X - E[X])^2]$$

(second central moment)

A Technical Remark

Not every random variable has a finite expectation.

Example: $E[1/X]$ where $X \sim \mathcal{U}(0, 1)$ $Y = 1/X$

For a continuous/discrete random variable, a sufficient condition is

$$E(|X|) = \int_{-\infty}^{\infty} |x| f_X(x) dx < \infty, \quad \sum_x |x| p_X(x) < \infty.$$

$-|x| \leq X \leq |x|$

Throughout most of this course, when we write $E[X]$, we will implicitly assume that the expectation is well defined and finite.

$$E(Y) = \int_0^1 1 \cdot \frac{1}{x} dx = \infty$$

$g(x) = 1/x$

Takeaway

Expectation measures the average.

Variance measures the spread of the random variable around its mean.

The most useful computational tool is

$$E[g(X)] = \int g(x) f_X(x) dx$$

for continuous random variables, or

$$E[g(X)] = \sum_k g(x_k) P_X(x_k)$$

for discrete ones.